Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic decisions

Made with GAMMA

Dataset Overview

3,900

Transactions

Purchase records analyzed

18

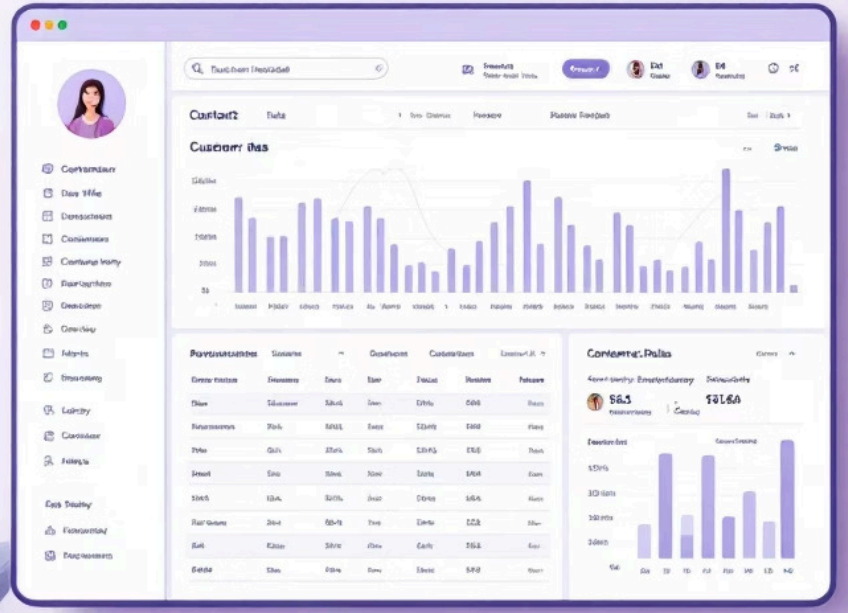
Features

Data points per transaction

37

Missing Values

Review ratings imputed



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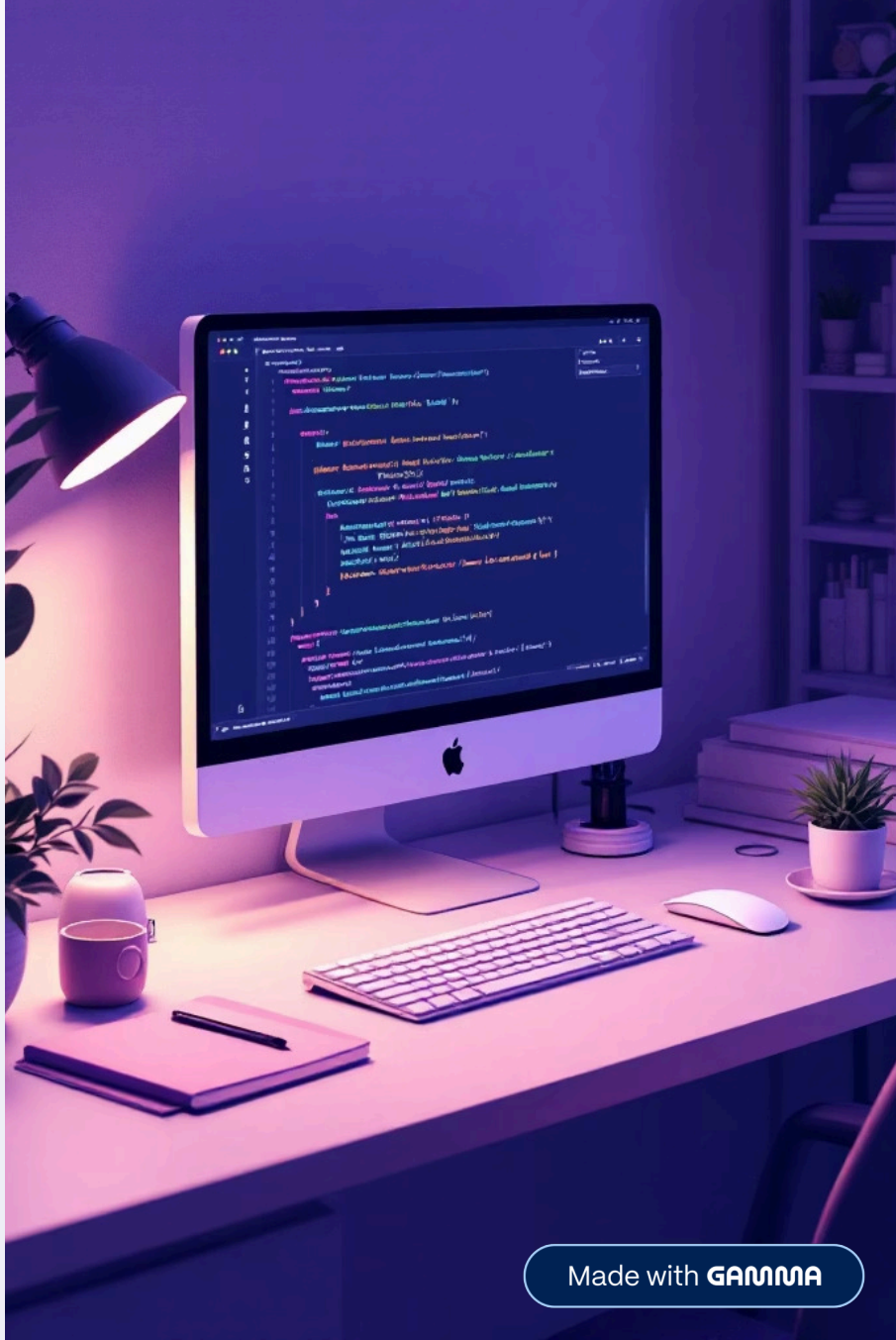
Python Data Preparation

Data Loading & Exploration

- Imported with `pandas`
- Checked structure with `df.info()`
- Summary statistics via `.describe()`

Cleaning & Engineering

- Imputed missing review ratings
- Standardized to snake case
- Created age groups and purchase frequency



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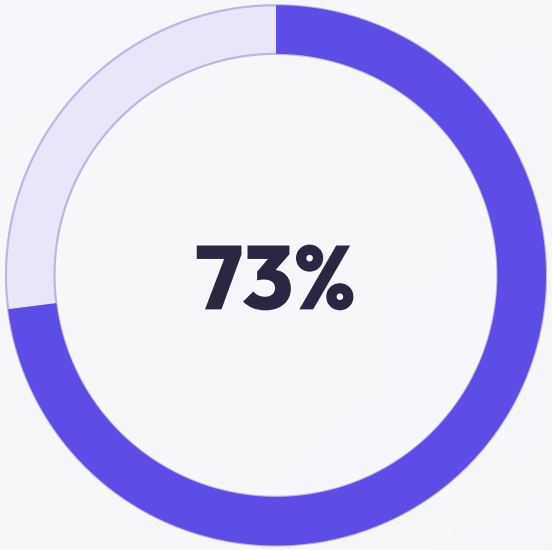
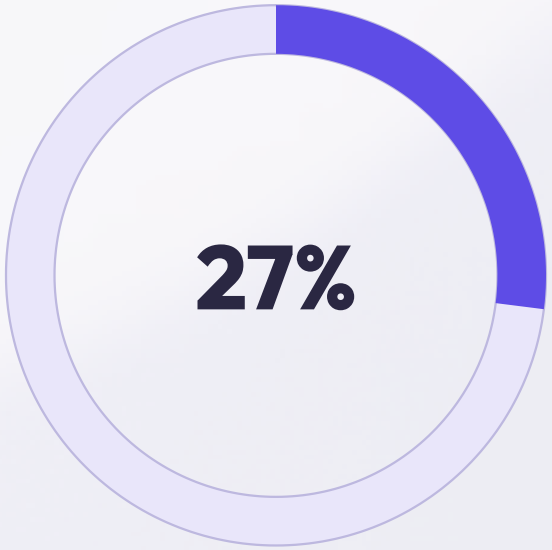
Revenue by Gender



Male Customers

$157,890 revenue



Female Customers

$75,191 revenue

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Product Performance

Top Rated

Gloves, Sandals, Boots

Most Discounted

Hat, Sneakers, Coat

Best Sellers

Blouse, Pants, Jewelry

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Shipping & Discount Analysis

Shipping Comparison

Express: **$60.48** avg

Standard: **$58.46** avg

High-Spending Discount Users

839 customers used discounts while spending above average

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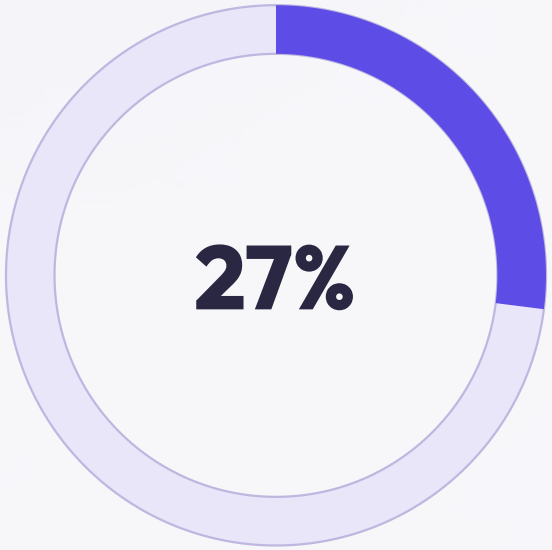
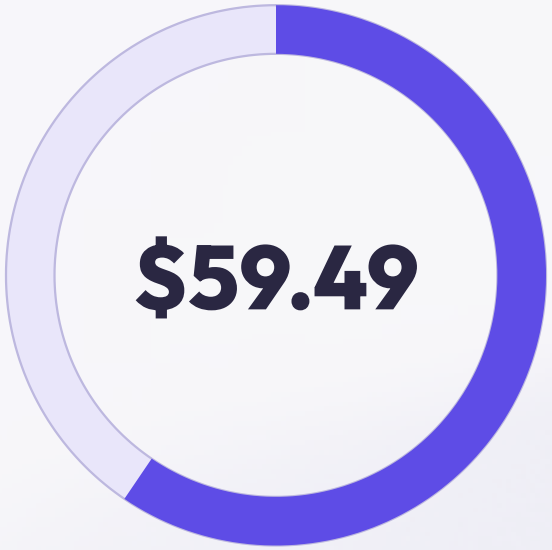
Subscription Insights



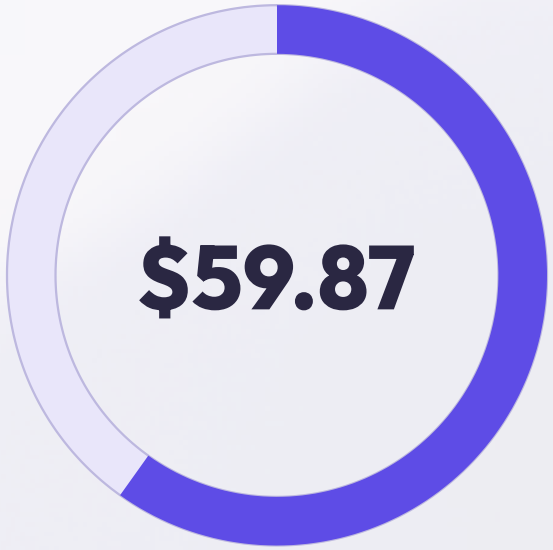
Subscribers

1,053 customers



Avg Spend

Subscribers



Avg Spend

Non-subscribers

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Customer Segmentation

1

Loyal

3,116 customers

2

Returning

701 customers

3

New

83 customers

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Revenue by Age Group

Young Adult

$62,143

Middle-aged

$59,197

Adult

$55,978

Senior

$55,763

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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